

Profit Analysis Project

Objective

The objective of this project is to analyze the impact of R&D, Marketing, and Administration spending on startup profitability using regression analysis and business intelligence visualization. The analysis supports data-driven decision making and profit optimization.

Dataset Overview

Source: MySQL Database (project_profit_analysis), Table: startup, 50 startup records, Columns: RD_Spend, Administration, Marketing_Spend, State, Profit. The dataset contains startup financial data from California, Florida, and New York.

Column	Type	Description	Role
RD Spend	Numerical	Amount invested in Research and Development	Independent Variable
Administration Spend	Numerical	Administrative operating expenses	Independent Variable
Marketing Spend	Numerical	Marketing and advertising expenditure	Independent Variable
State	Categorical	Location of the startup (California, Florida, New York)	Categorical Predictor
Profit	Numerical	Net profit earned by the startup	Target Variable

The dataset contains four predictor variables and one target variable. Profit is the dependent variable because it is the value being predicted, whereas the remaining variables are used to explain variations in profit.

Data Cleaning

The dataset was preprocessed before building the regression model using the following steps:

Missing Value Detection

- Each column was examined for NULL or missing values using SQL queries and Orange Data Table.
- No missing values were found.

Duplicate Record Validation

- Duplicate startup records were checked.
- No duplicate observations were identified.

Data Type Verification

- RD_Spend, Administration, Marketing_Spend, and Profit were verified as numerical variables.
- State was verified as a categorical variable.

Categorical Variable Handling

- The categorical variable **State** cannot be used directly in linear regression.
- Orange internally converted **State** into **numeric dummy / one-hot encoded indicators** before model training.
- This allowed the model to treat **California, Florida, and New York** as separate location signals without losing information.

Range Validation

- Minimum and maximum values of each numerical attribute were reviewed to detect invalid entries.

Outlier Assessment

- Scatter plots and summary statistics were examined.
- No extreme outliers requiring removal were observed.

Final Dataset Validation

- The dataset was confirmed to contain 50 complete observations and was suitable for regression analysis.

Analysis

Linear Regression was used with Profit as the target variable. Independent variables included RD_Spend, Administration, Marketing_Spend, and State. Metrics such as R^2 , MSE, RMSE, and MAE were evaluated. The model achieved approximately $R^2 = 0.944$, indicating strong predictive capability.

Regression Assumptions

The Multiple Linear Regression model was developed based on the following assumptions:

- A linear relationship exists between the predictor variables and Profit.
- Each startup record represents an independent observation.
- Prediction errors are randomly distributed around the regression line.
- Predictor variables are not highly correlated with one another.
- The variance of prediction errors remains approximately constant across different observations.

Model Limitations

Although the regression model achieved an R^2 value of approximately **0.944**, several limitations should be considered.

- The dataset contains only **50 startup observations**, which limits the model's generalization ability.
- The analysis includes startups from only **three states** (California, Florida, and New York).
- External business factors such as market competition, customer demand, inflation, taxation, and economic conditions were not included in the model.

Therefore, the regression model should be used as a **decision-support tool** to estimate startup profitability. Final investment decisions should also consider external business and economic factors that are not included in the model.

Future Enhancement Note

A more advanced version of this analysis could include feature selection or regularized regression such as Ridge or Lasso. These methods can help improve stability, reduce overfitting risk, and test whether a simpler model performs better on larger datasets.

Figure 1: Startup dataset extracted and validated from the MySQL database.

REGRESSION ANALYSIS FLOWCHART

A Linear Regression model was developed in Orange to analyze the relationship between startup expenditures and profit. The model was trained using R&D Spend, Marketing Spend, Administration Spend, and State as predictor variables, while Profit was used as the target variable.

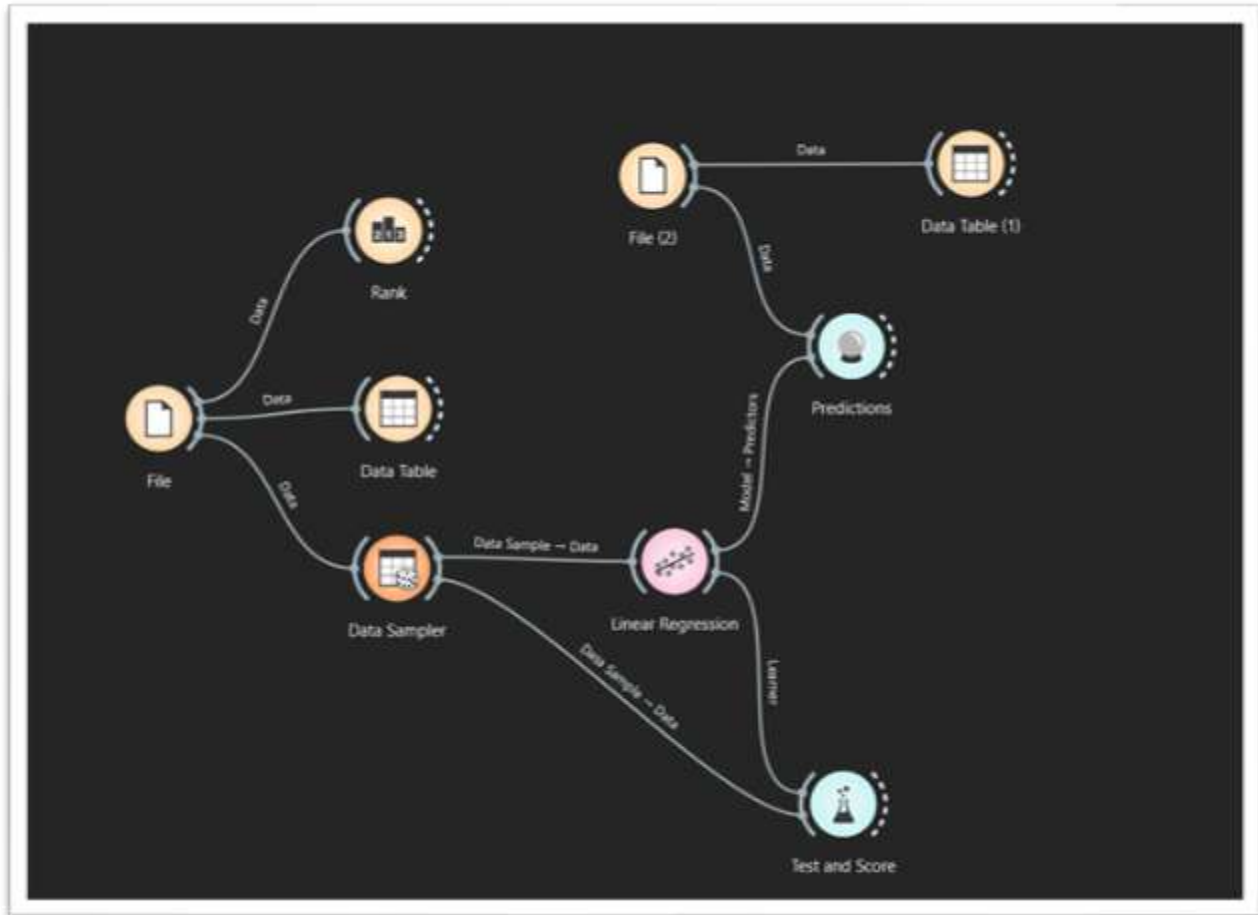


Figure 2: Linear regression workflow developed in Orange for profit prediction.

FEATURE IMPORTANCE ANALYSIS

Feature ranking was performed to evaluate the relative influence of different variables. The results indicate that spending-related variables have a significant impact on profit, helping identify the most important factors affecting business performance.

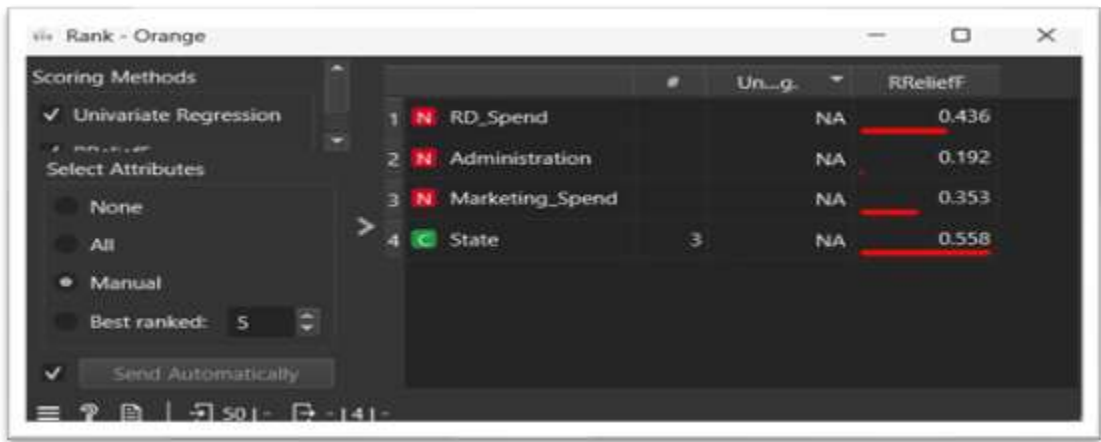


Figure 3: Feature ranking analysis showing the relative importance of variables in influencing startup profit.

Interpretation of Feature Ranking

The feature ranking analysis indicates that R&D Spend is the most influential numerical factor affecting startup profit. This suggests that organizations investing more in research and development generally achieve higher profitability.

Marketing Spend also contributes positively toward profit generation; however, its influence is comparatively lower than R&D investment.

Administration demonstrates the weakest influence among the spending variables, indicating that increasing administrative expenditure alone is unlikely to produce significant improvements in profitability.

The ranking results are consistent with the regression model and scatter plot analysis, providing additional confidence in the overall findings.

MODEL PERFORMANCE EVALUATION

The regression model achieved an R^2 value of approximately 0.944, indicating that about 94.4% of the variation in profit can be explained by the selected variables. This demonstrates strong predictive performance and reliability.

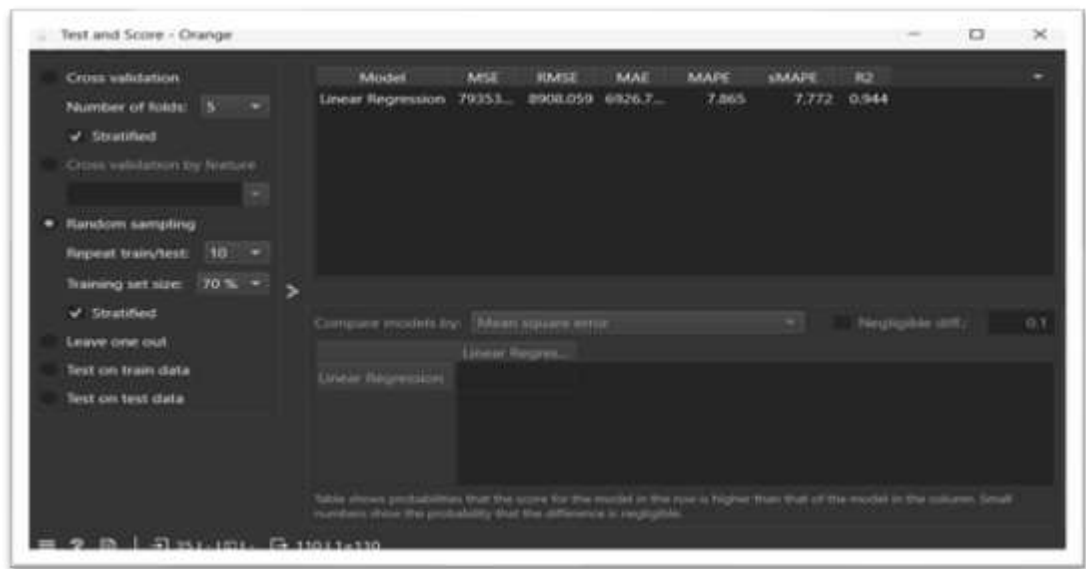


Figure 4: Regression model evaluation showing prediction accuracy and performance metrics.

PROFIT FORECASTING RESULT

The trained regression model was used to predict profit for two new startup scenarios provided in the project requirements. These predictions demonstrate the practical application of the model for future business planning and investment decisions.

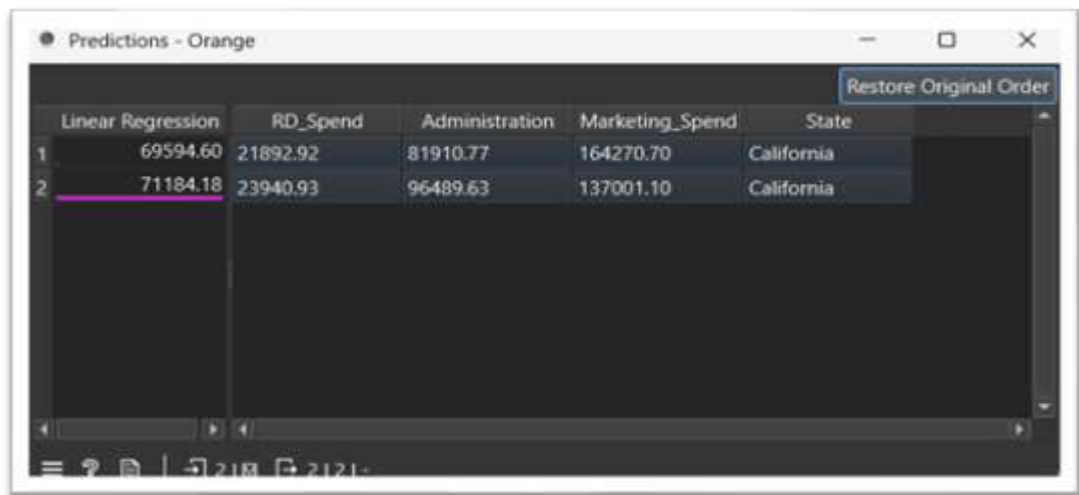


Figure 5: Predicted profit values generated for the given startup investment scenarios.

Predicted Values of the Profit for the below-given input features are:

R&D SPEND	ADMINISTRATION SPEND	MARKETING SPEND	PROFIT
21892.92	81910.77	164270.7	69594.60
23940.93	96489.63	137001.1	71184.18

BUSINESS INTELLIGENCE DASHBOARD OVERVIEW

An interactive Power BI dashboard was developed to visualize key business metrics, spending patterns, and profitability trends. The dashboard enables quick exploration of insights and supports data-driven decision-making.

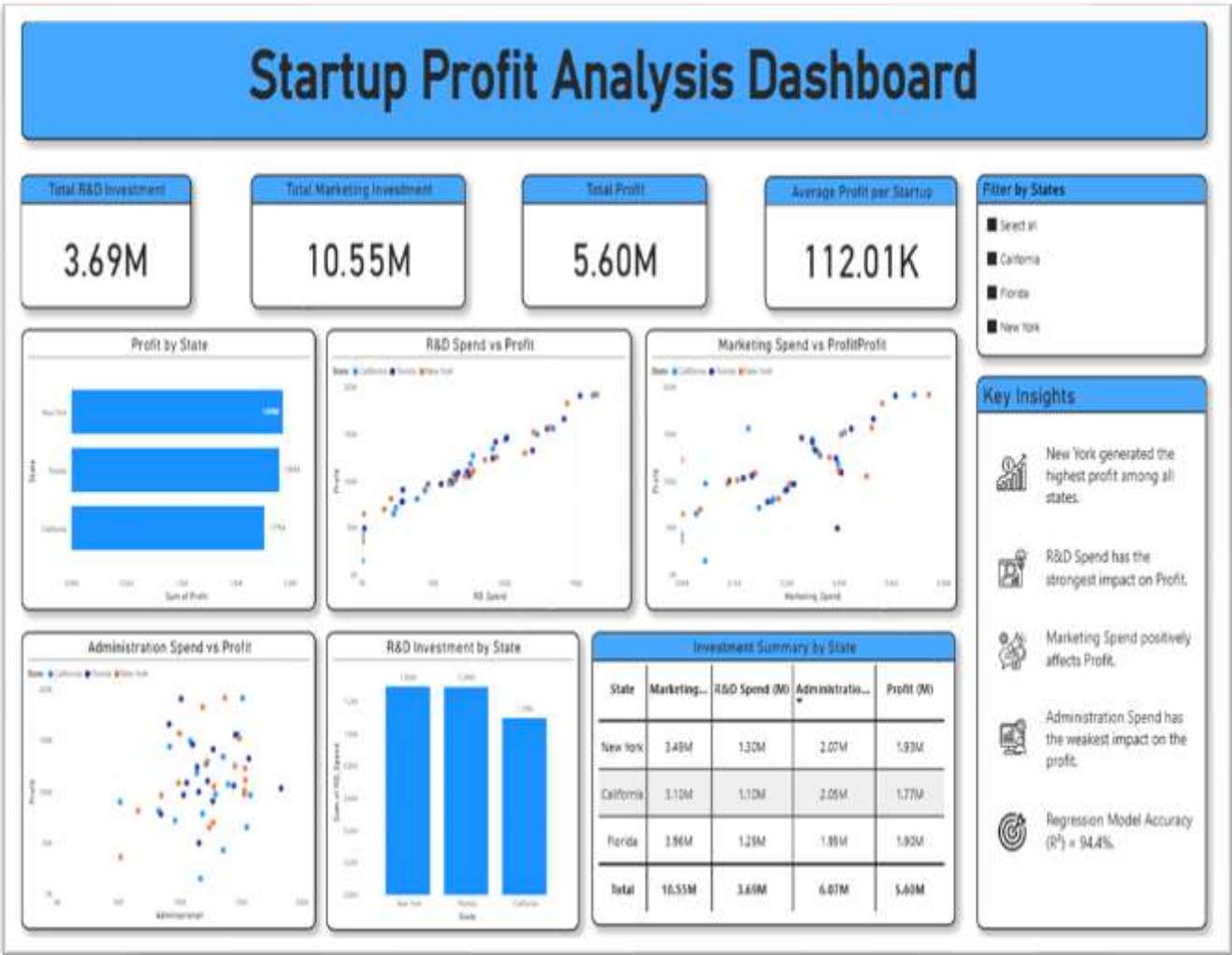


Figure 6: Interactive Power BI dashboard summarizing startup profit analysis.

STATE-WISE PROFIT ANALYSIS

This visualization compares total profit generated across different states. New York recorded the highest profit, followed by Florida and California, indicating regional differences in startup performance.

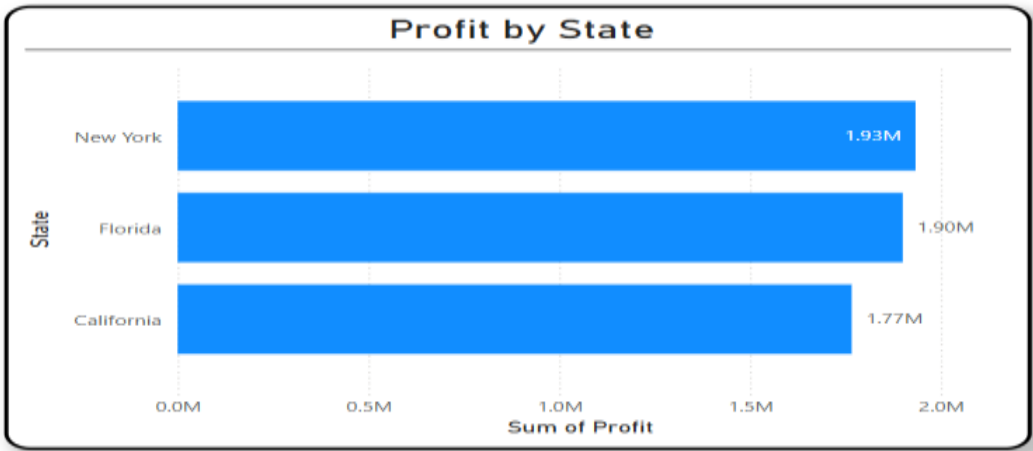


Figure 7: Comparison of total profit generated across different states.

IMPACT R&D INVESTMENT ON PROFITABILITY

The scatter plot shows a strong positive relationship between R&D spending and profit. Startups investing more in research and development generally achieved higher profitability, making R&D the strongest profit driver.

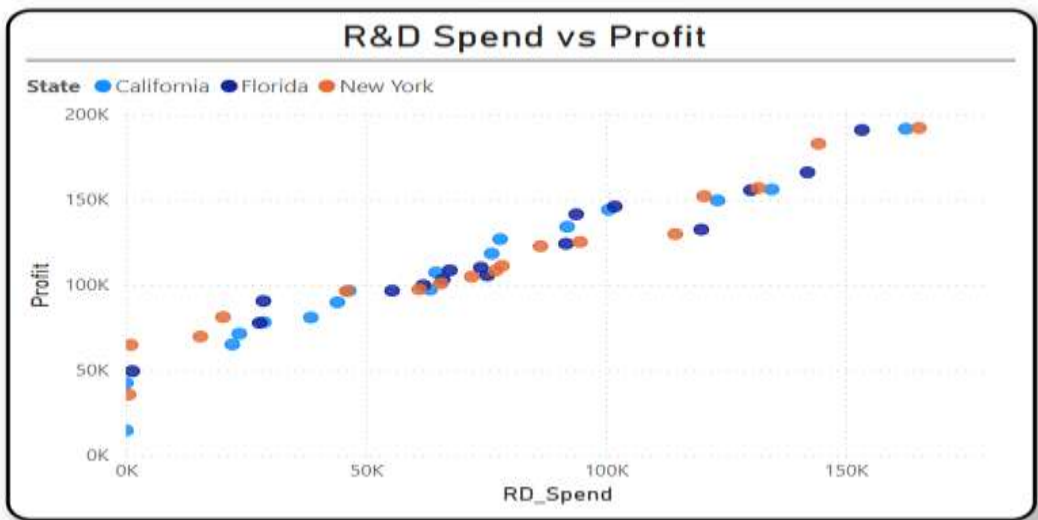


Figure 8: Relationship between R&D investment and startup profitability.

IMPACT MARKETING INVESTMENT ON PROFITABILITY

The analysis indicates a positive relationship between marketing expenditure and profit. While the relationship is not as strong as R&D spending, increased marketing investment is generally associated with improved profitability.

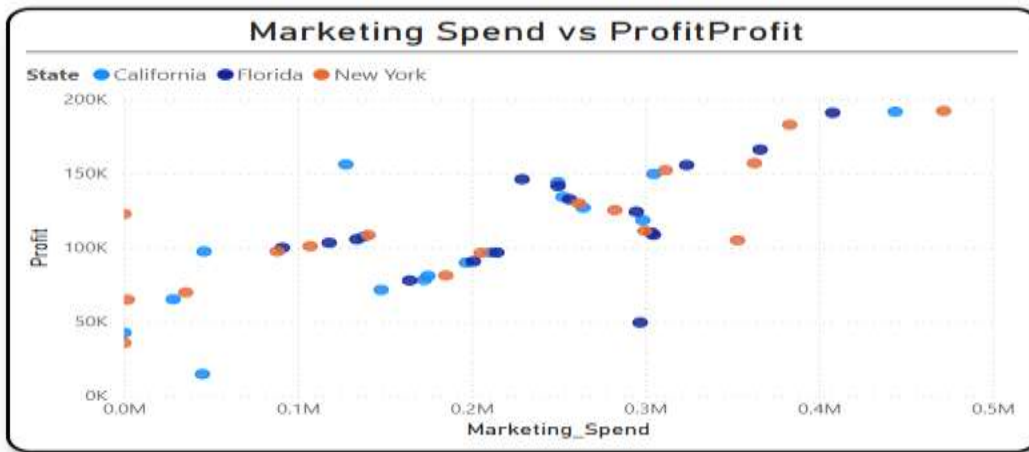


Figure 9: Impact of marketing expenditure on startup profit.

IMPACT ADMINISTRATION INVESTMENT ON PROFITABILITY

The scatter plot reveals a relatively weak relationship between administration spending and profit. This suggests that higher administrative expenses do not necessarily translate into higher profitability.

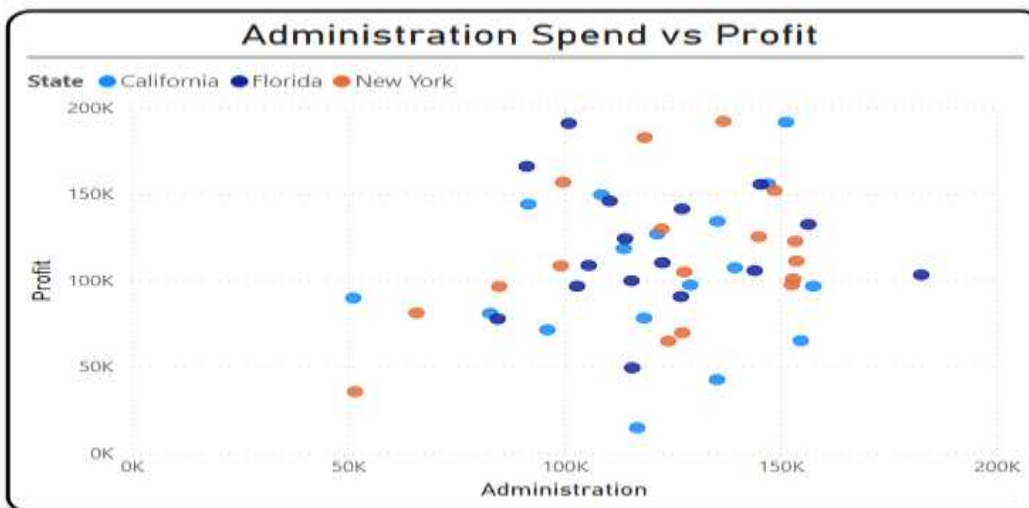


Figure 10: Analysis of administration expenditure and its relationship with profit.

STATE-WISE R&D INVESTMENT COMPARISON

The chart compares total R&D investment across New York, Florida, and California. New York and Florida recorded the highest R&D spending, while California invested comparatively less. This pattern aligns with the profitability analysis, suggesting that higher investment in research and development may contribute to improved business performance and profitability.

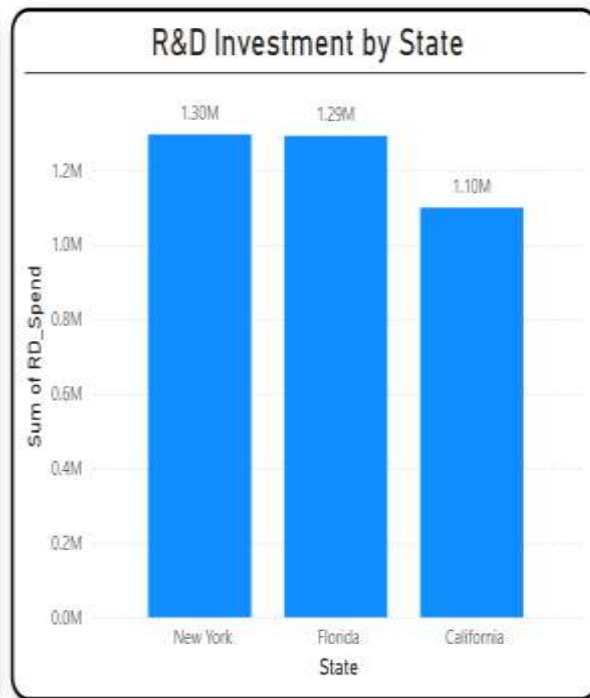


Figure 11: Comparison of R&D investment levels across states.

Insights

- **New York generated the highest cumulative profit** among the three states, indicating stronger overall startup financial performance within the available dataset.
- **R&D Spend exhibited the strongest positive relationship with Profit.** Startups investing more in research and development consistently achieved higher profit, making R&D the most influential investment category.
- **Marketing Spend showed a moderate positive relationship with Profit.** Although its impact was lower than R&D, increased marketing expenditure generally contributed to improved profitability.

- **Administration Spend displayed the weakest relationship with Profit.** Higher administrative expenses did not consistently result in increased profit, suggesting limited financial returns from additional administrative spending.
- **The regression model achieved an R^2 value of approximately 0.944**, indicating that about **94.4% of the variation in startup profit** can be explained by the selected predictor variables.
- **The prediction results demonstrated the practical usefulness of the regression model**, allowing startup profit to be estimated for new investment scenarios before actual business decisions are made.
- **The Power BI dashboard enabled interactive visualization of business performance**, allowing stakeholders to compare spending patterns, regional performance, and profitability trends efficiently.

Recommendations

- **Priority 1: Increase R&D investment** because it showed the strongest positive impact on profit.
- **Priority 2: Maintain marketing investment** to support growth, but track return carefully before scaling further.
- **Priority 3: Review and optimize administration expenses** because they showed the weakest profit impact.
- **Use the regression model for budget planning** before making future spending decisions.
- **Benchmark New York startup practices** since New York recorded the highest cumulative profit.
- **Do not blindly increase spending in every area**; more spending only helps when it produces measurable profit improvement.
- **Run ROI checks for major budget changes** so the company can prioritize money where it gives the highest return.

Conclusion

The project successfully identified major profit drivers, developed an accurate regression model, and produced actionable business insights through Power BI visualization.

The integration of SQL, Orange, and Power BI demonstrates how data engineering, predictive analytics, and business intelligence can be combined into a single analytical workflow. The proposed framework can support management teams in making evidence-based investment decisions and optimizing resource allocation.